Chapter 3
Medium Access

Lecture Mobile Communication

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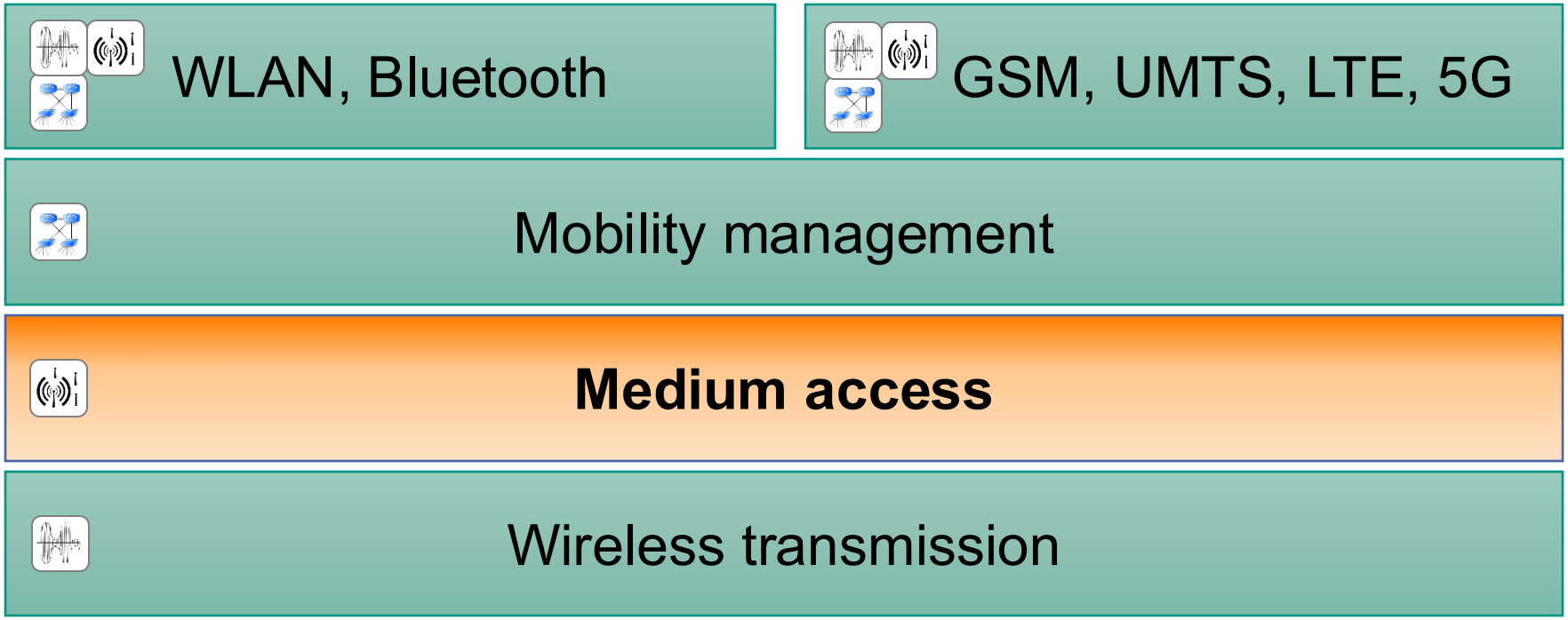
Structure


WLAN, Bluetooth
GSM, UMTS, LTE, 5G
Mobility management
Medium access
Wireless transmission

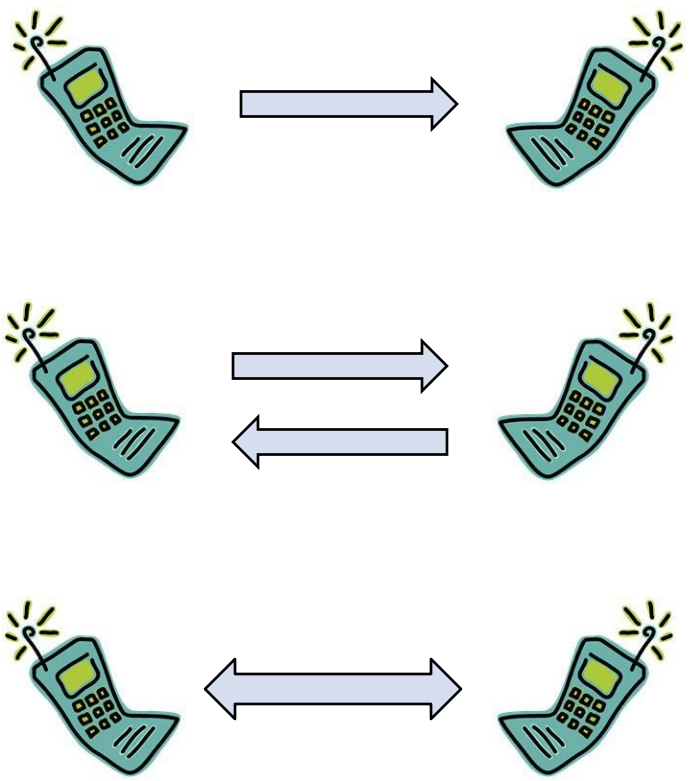
Introduction

Fundamental problem

- Several stations compete for access to the same shared resource (medium)
- The obvious solution: multiplexing (see Chapter 2)
  - Simultaneous use of a medium by multiple channels
  - Possible in four dimensions
    - Room (r),
    - Time (t),
    - Frequency (f) and
    - Code (c)

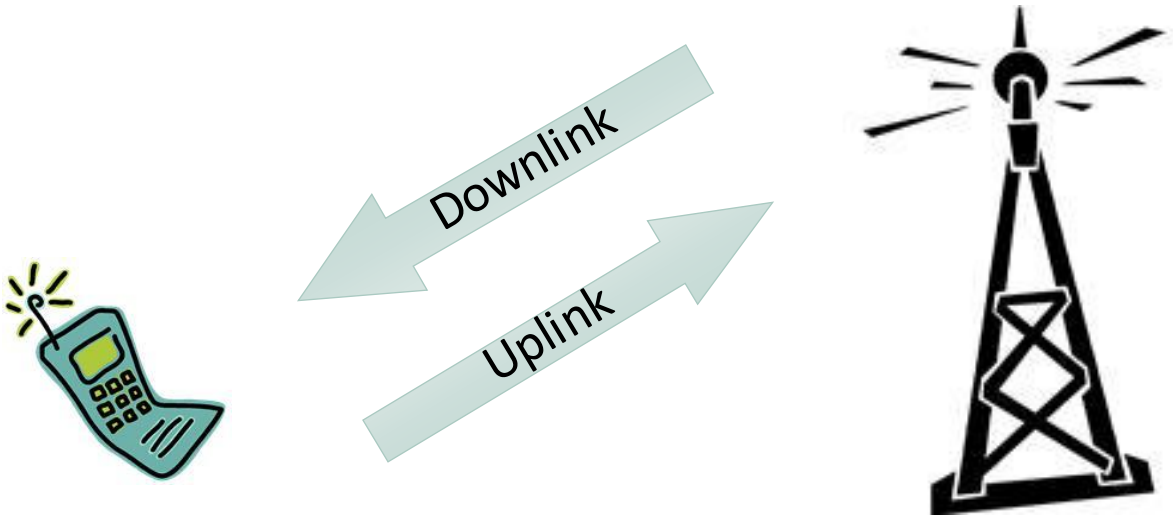
Example: Point-to-Point Communication

- Two stations communicate with each other over a shared medium
  - Obviously, a multiple access problem!
- Communication direction
  - Simplex
    - Transmission in one direction
  - Half-duplex
    - Transmission in both directions
    - Not at the same time
  - (Full) duplex
    - Transmission in both directions
    - At the same time



Duplex Communication

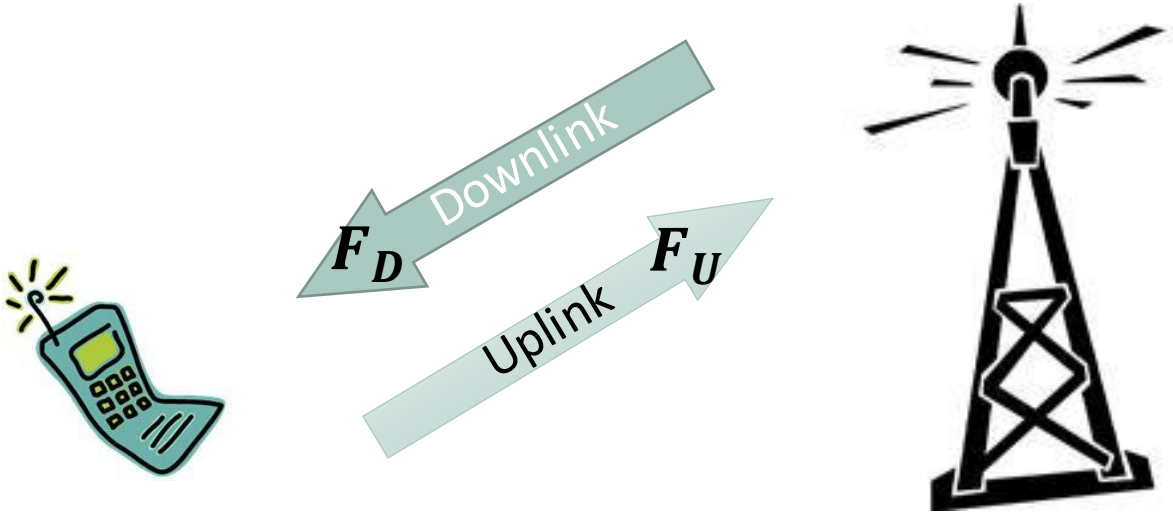
- Often considered on the last wireless link between network and user device
- Uplink
  - Communication from the user device to the network
- Downlink
  - Communication from the network to the user device



Duplex communication

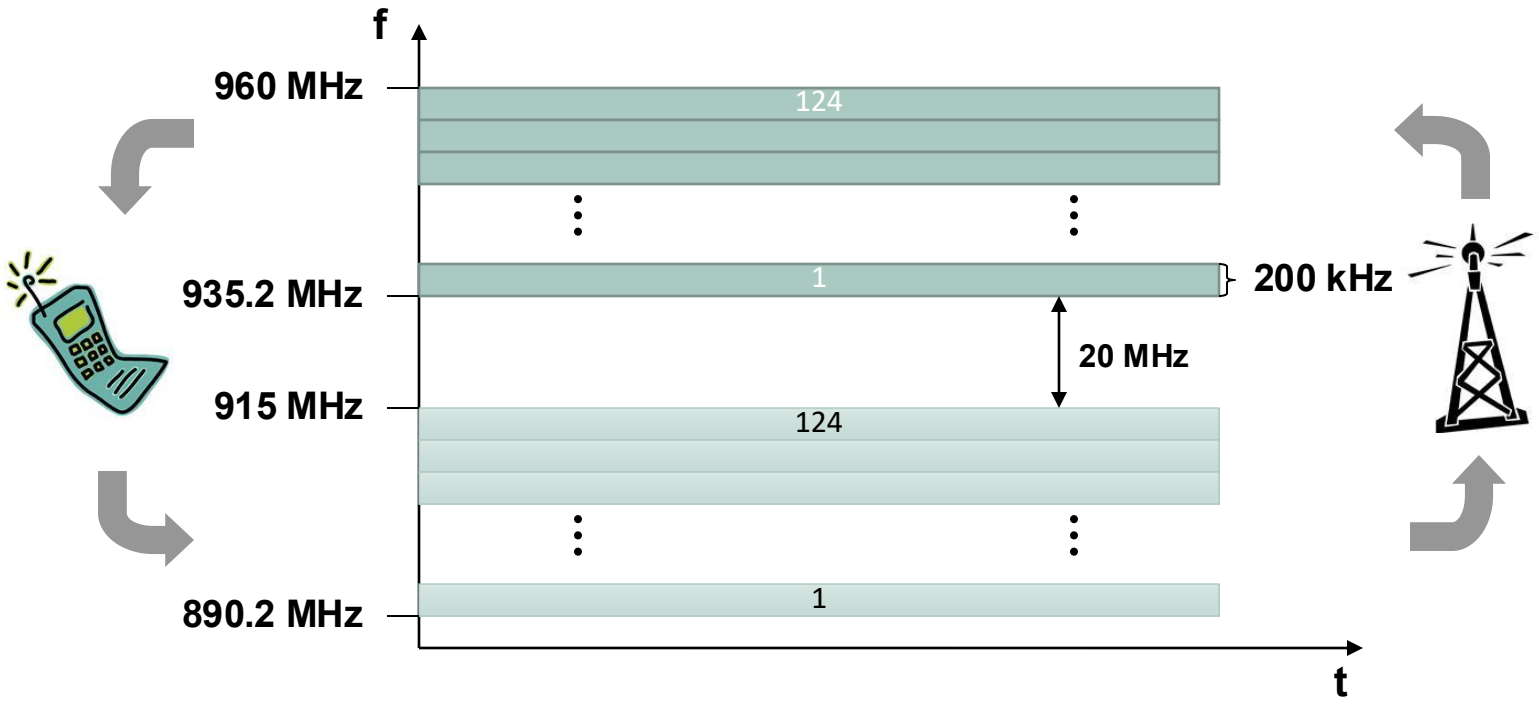
- Frequency Division Duplex (FDD)
  - Build on frequency division multiplex
  - Different frequencies for up-/downlink
- Time Division Duplex (TDD)
  - Build on time division multiplex
  - Different times for up-/downlink
    - Up- and downlink alternate quickly

Frequency Division Duplex

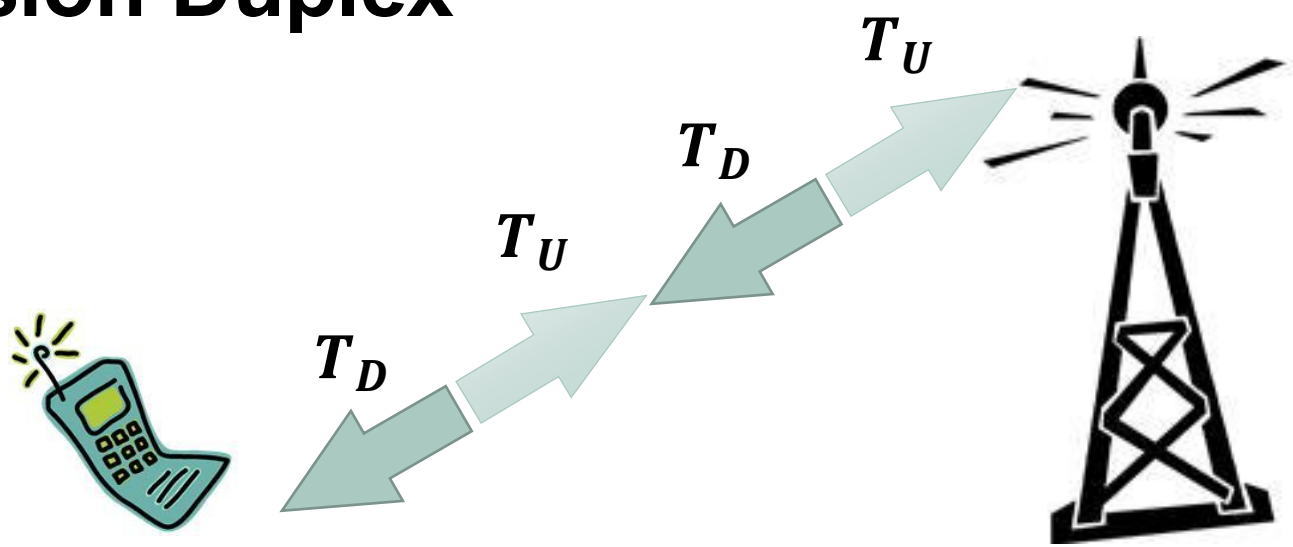


- Separate frequencies for uplink and downlink
  - + Interference for uplink and downlink independent
  - - No (simple) support for asymmetric data traffic
- Example:
  - LTE (FDD), UMTS (FDD), GSM, IS-95, AMPS

Example: FDD in GSM

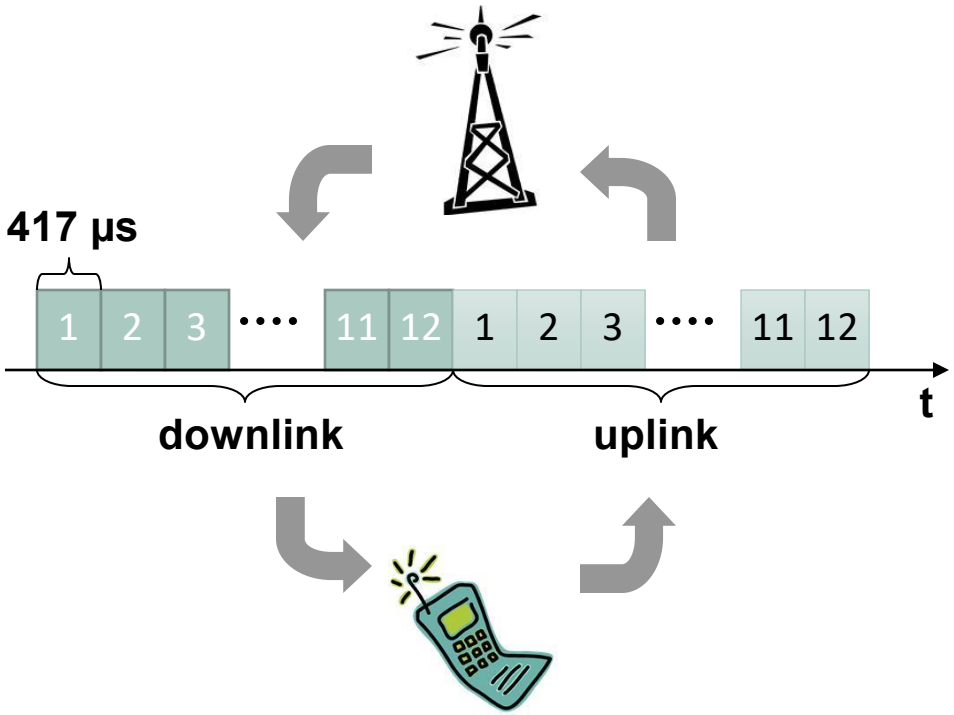


Time Division Duplex



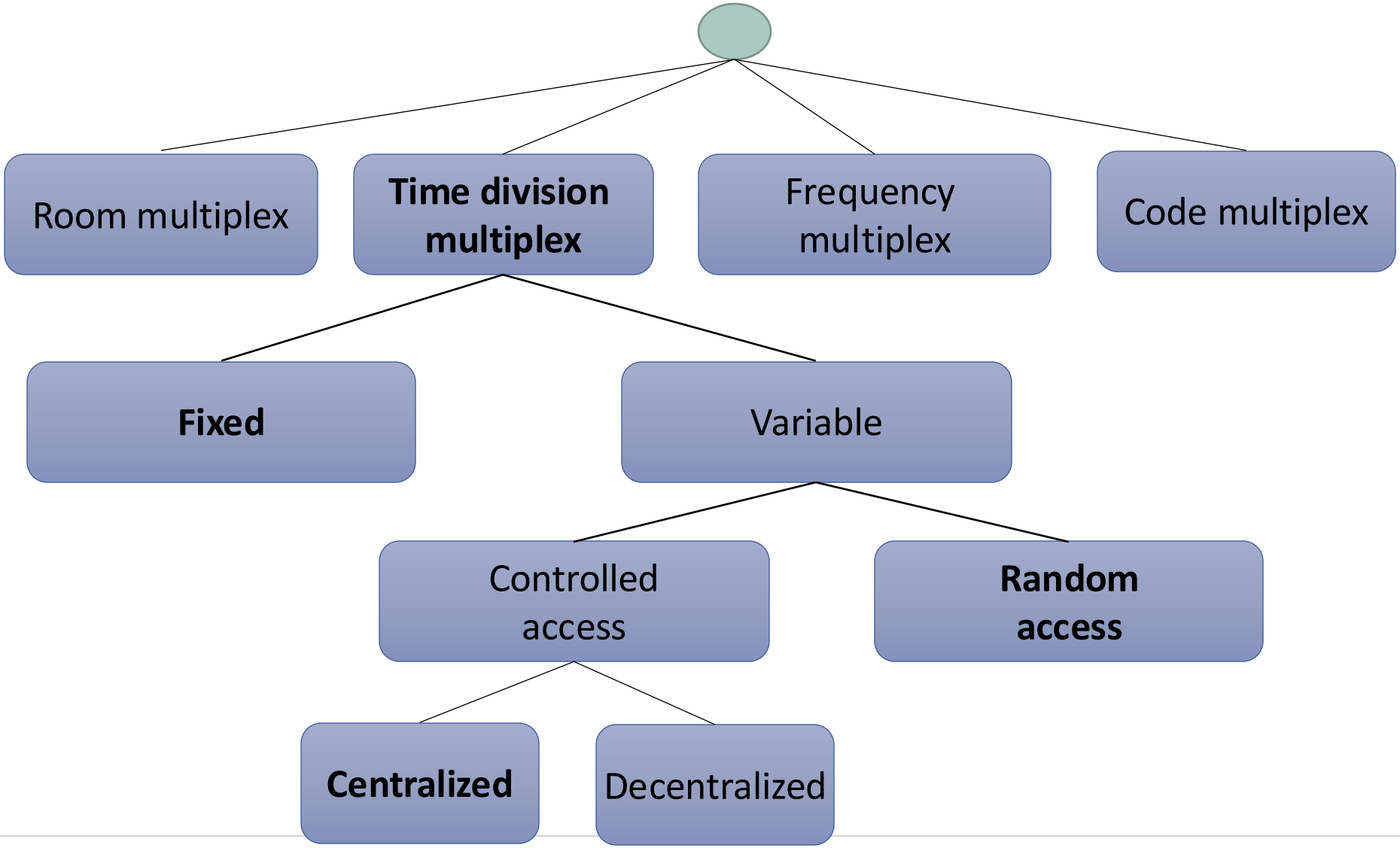
- Separate time slots for uplink and downlink
  - + Support for asymmetric data traffic (length of time slots)
  - - Mixing of interference on uplink and downlink
- Examples
  - DECT, WIFI, LTE (TDD), UMTS (TDD)

Example: TDD in DECT

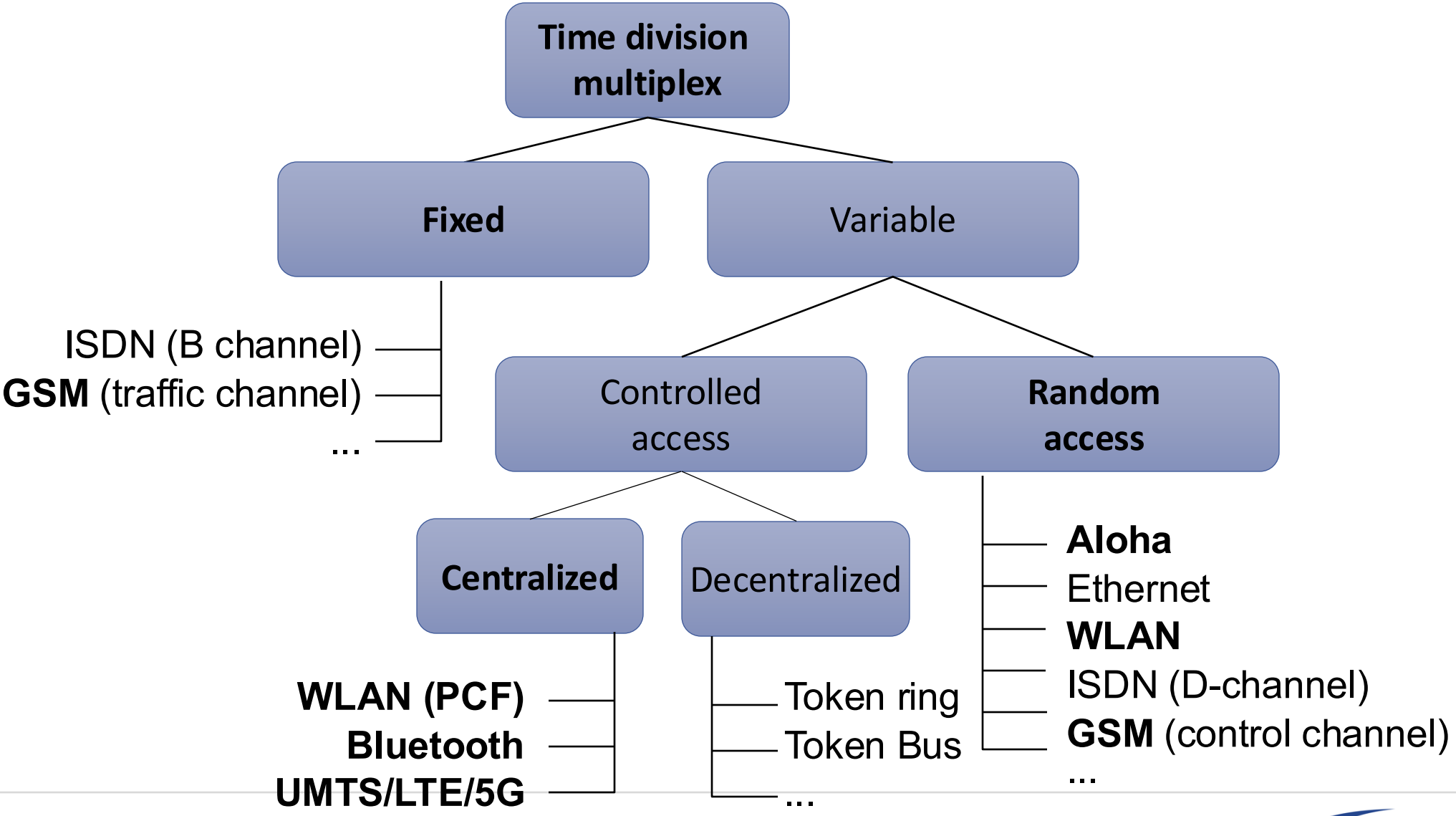


- Multiple use of time division multiplex
  - Time duplex (uplink, downlink)
  - Fixed assignment of time slots in uplink and downlink

Medium Access - Categories

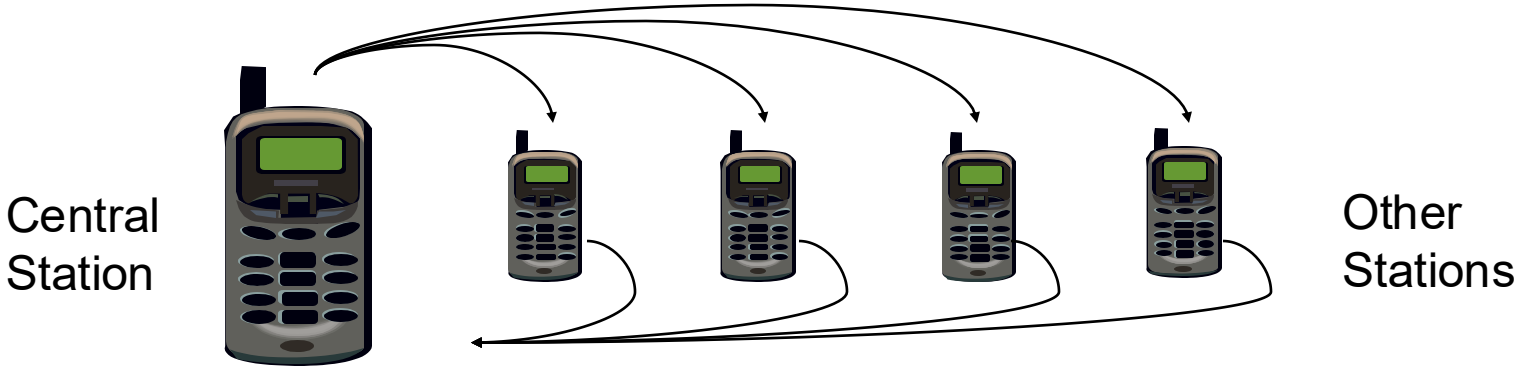


Media access - example networks



Options for Controlled Centralized Access

- Control via central station (polling)
  - Queries other stations according to polling table
  - Devices only respond when polled
- Disadvantages
  - Central device may fail
  - Load on central device may become high



Options for Random Access

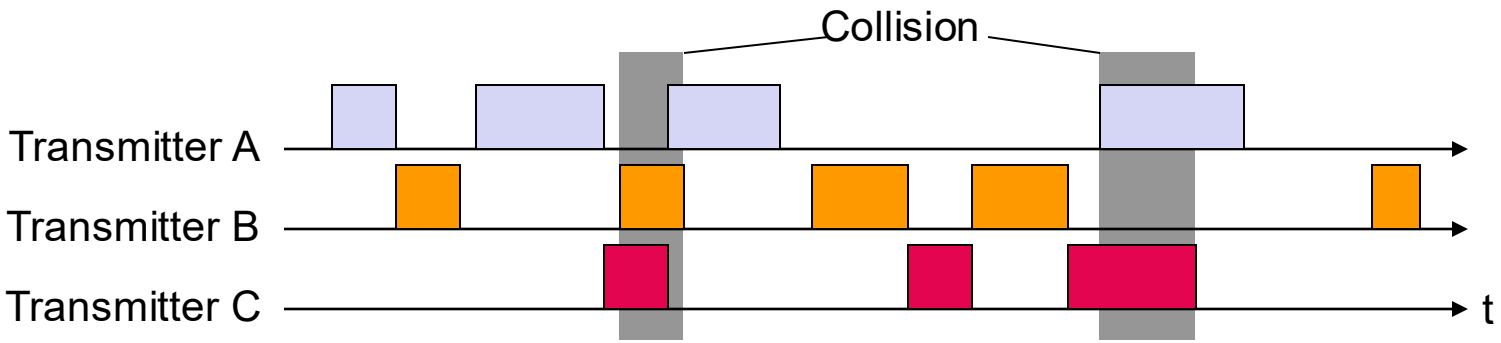
- Timing
  - Asynchronous
    - Access to the channel possible at any time
    - Examples
      - Aloha, Ethernet, Token Ring
  - Synchronous
    - Division of the channel into time slots
      - Access at the beginning of each time slot
      - → All stations must be synchronized in time
    - Examples
      - Slotted Aloha, ISDN B-channel

- Carrier Sense - Listen before talk?
  - Useful when propagation delay ≪ transmission time

Random Access without Carrier Sense

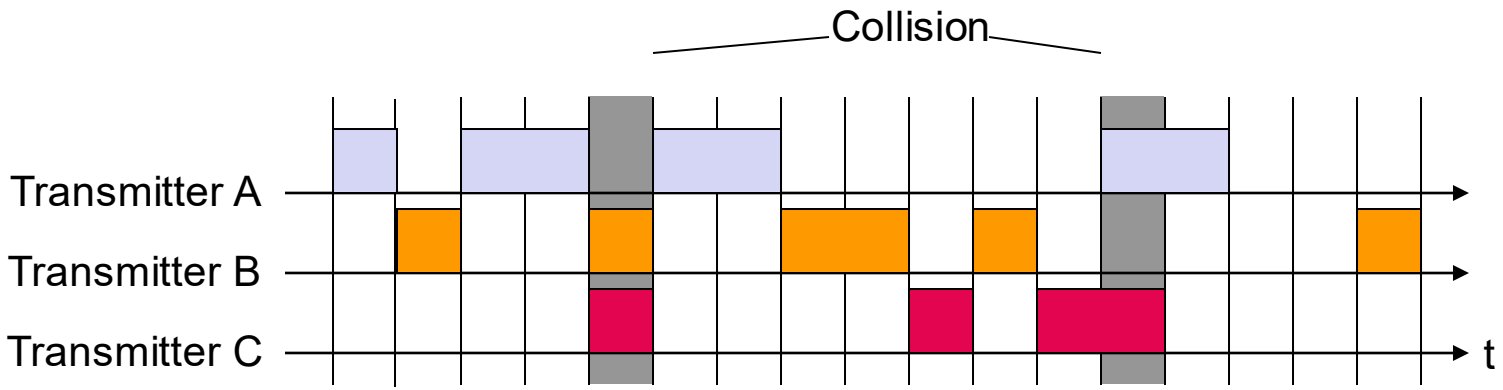
Aloha

- The first MAC protocol for packet-based *wireless* networks
  - Media allocation
    - Time-division multiplex, variable, random access
    - No (!) carrier sense
    - Asynchronous access
  - Problem
    - Collisions possible
    - Collision detection: Confirmations by higher layers necessary
- Scheme



Slotted Aloha

- Like Aloha, <u>but</u> additional use of time slots
  - <u>Synchronous</u> access – only at the start of a time slot
  - On average fewer collisions than Aloha
- Example of use
  - GSM, UMTS, LTE (control channel "Random Access Channel")



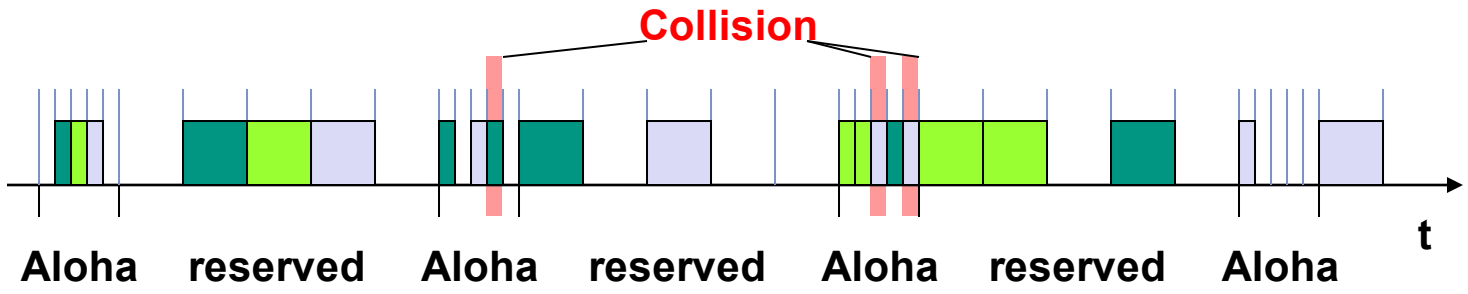
Explicit reservation

- Goal
  - Improvement of Aloha through advance reservations
    ... also known as Reservation Aloha
  - Combines random access with fixed / controlled access

- Distinction between two phases
  - Reservation phase
    - Variable, random access to (short) reservation time slots
      - Synchronous (Slotted) Aloha
    - Collisions possible
  - Data transfer phase
    - Fixed / controlled access to previously explicitly reserved data timeslots
    - No collisions possible
    - No data can be sent without explicit reservation of the corresponding data timeslot
      → Explicit reservation procedure

Example: Explicit reservation

- Scheme

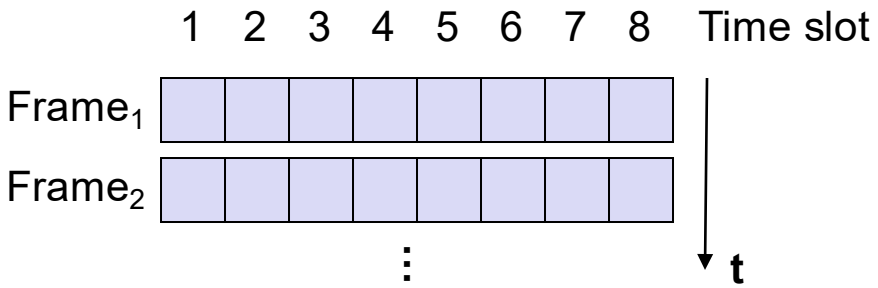


- Problem
  - All stations must have a consistent view of reservations!

- Example:
  - Dedicated data channels in GSM, UMTS, LTE – reserved in advance on random access channel implementing slotted aloha

Example: Implicit reservation

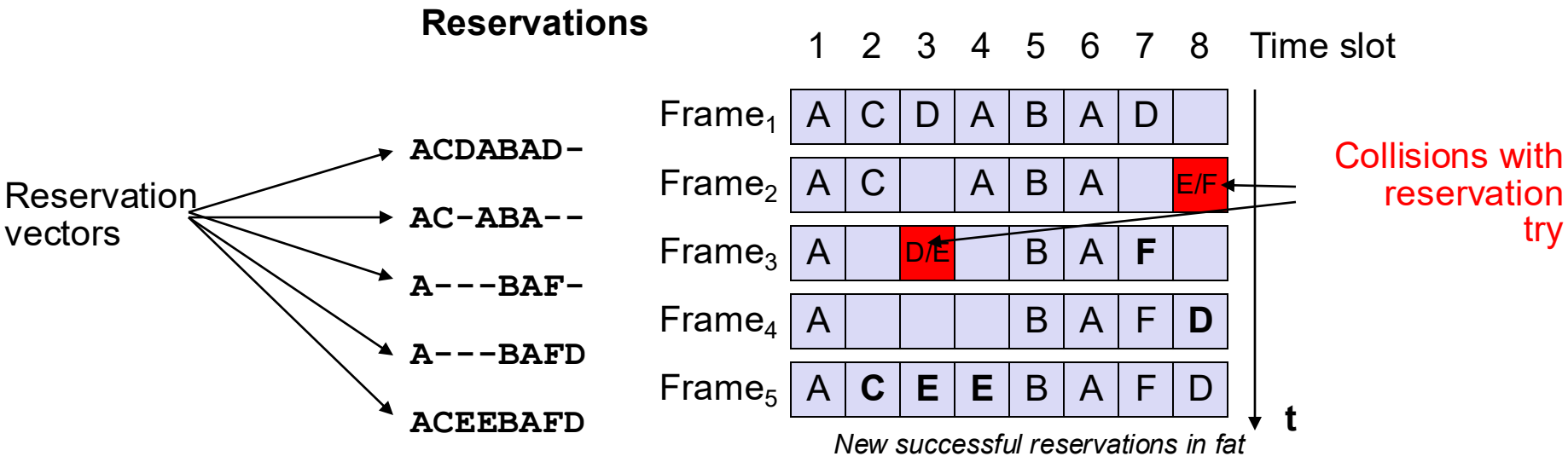
- Goal
  - Reservation over a longer period without an explicit reservation message
- Communication structure
  - Numbered time slots organized in repeating frames



- Approach
  - A timeslot used by a station in one frame can only be used by this station in the next frame → Timeslot is reserved
  - If reservation is no longer required, station leaves the timeslot unused
  - If time slot was unused in one frame, it can be reserved in the next frame by sending a data unit
    - Collision possible → (Slotted) Aloha, optional backoff
- Example:
  - WLAN - reservation is lost when the medium is unused for a certain time (Distributed Coordination Function Interframe Space, DIFS)

Example: Implicit reservation

- Assumptions:
  - Frames with 8 time slots
  - No backoff after colision
  - All stations know state of reservation (reservation vector) after a frame



Example: Implicit reservation (explanation)

- Procedure
  - All stations know reservation vector
    - E.g. "`AC-ABA--`"
    - Transmitted by base station or determined by carrier sensing
  - Free time slots in reservation vector indicate transmission opportunities
    - Variable, random access, synchronous (Slotted Aloha)
    - In case of collision, new attempt in later frame
      - Collision can be detected in the following reservation vector
  - Successful station keeps reservation as long as data is available for transmission
    - Time slots in the following frames can be used without explicit reservation

    → Implicit reservation procedure

Example: Implicit reservation (explanation)

- Explanation of example
  - Stations (here A-F) compete for free time slots
    - Frame 2
      - Station A continues using reserved time slots 1, 4, 6
      - Station B continues using reserved time slot 5
      - Station C continues using time slot 2
      - Station D releases reserved time slots 3 and 7
      - Collision of stations E and F in free time slot 8

Implicit reservation (explanation)

- Frame 3
  - Station A continues using reserved time slots 1, 6
  - Station B continues using reserved time slot 5
  - Station A releases reserved time slot 4
  - Station C releases reserved time slot 2
  - Collision of stations D and E in free time slot 3
  - Successful reservation of station F in free time slot 7
- Frame 4
  - Station A continues using reserved time slots 1, 6
  - Station B continues using reserved time slot 5
  - Station F continues using reserved time slot 7
  - Successful reservation of station D
- etc.

Random Access with Carrier Sense

Carrier Sense Multiple Access (CSMA)

- Carriers Sense:
  Transmitter scans medium before the start of transmission
  - Checks energy level
  - Media is only accessed if the medium is currently idle
  - Problem: Transmitter only has a purely local view, causing hidden and exposed terminals (see Chapter 1)
  - Hidden terminals may cause collisions

- Approaches to collision avoidance (CA)
  - Out-of-band signaling
  - In-band signaling

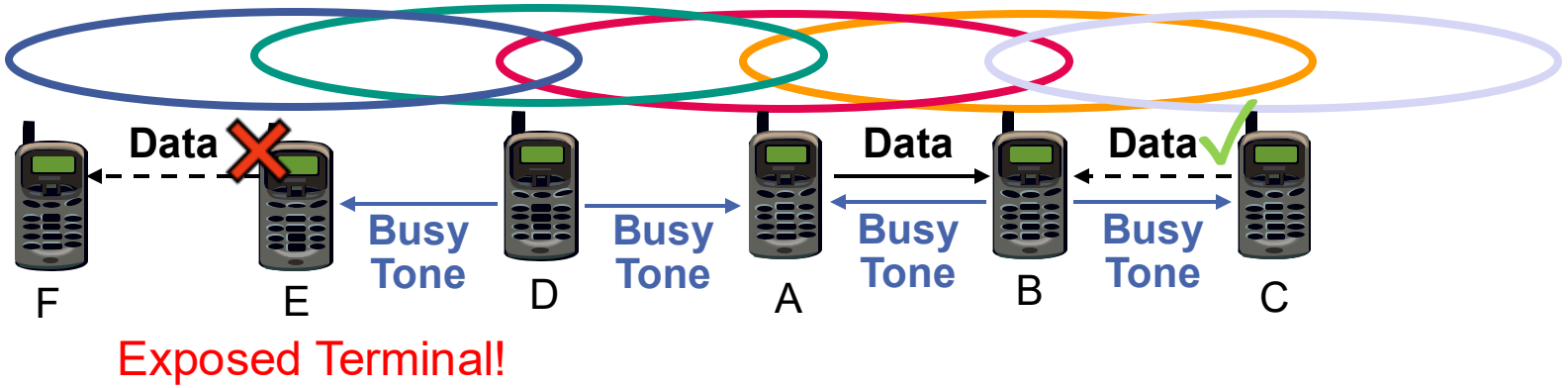
Approaches for Collision Avoidance (1)

- Collision avoidance using out-of-band signaling
  - Control and data messages separated by frequency multiplexing
    - Separate frequency bands for control channel and data channel
  - Advantage: No self-interference
  - Disadvantage: An additional frequency band must be assigned

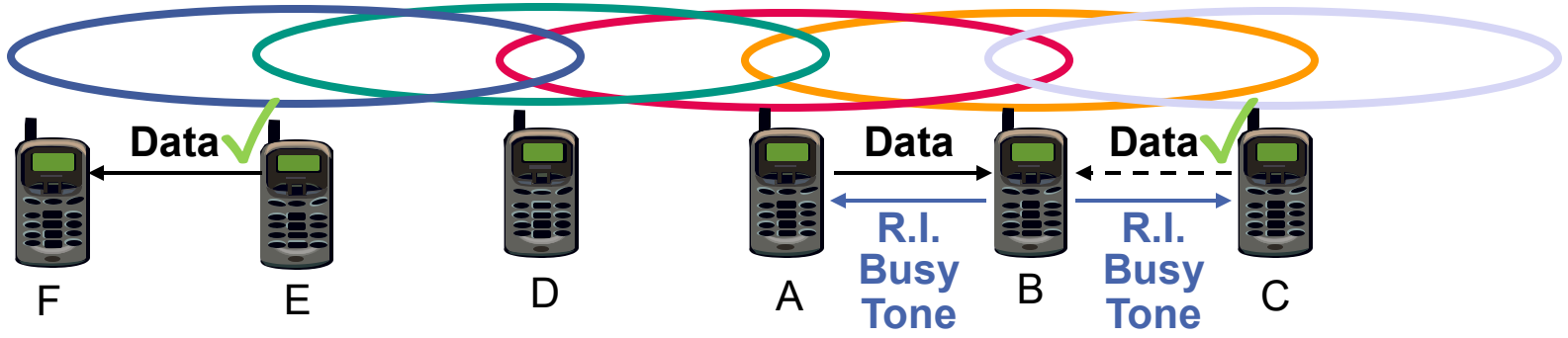
- Example
  - Busy Tone Multiple Access (BTMA)
    - Any station detecting a continuous transmission on data channel sends Busy Tone on control channel
    - Stations receiving a busy tone defer transmission (all devices within a 2-hop radius of an active station)
    - No hidden stations, but many exposed stations
  - Receiver Initiated Busy Tone Multiple Access (RI-BTMA)
    - Only intended receiver of a transmission sends Busy Tone
    - Fewer exposed stations, but busy tone can only be sent after receiver has decoded transmission request

Example: BTMA vs. RI-BTMA

- Busy Tone Multiple Access



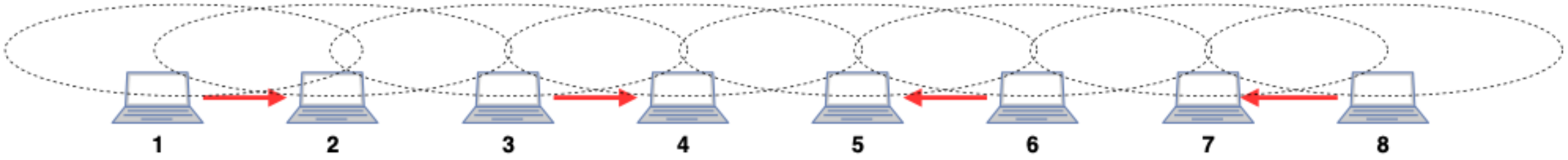
- Receiver Initiated Busy Tone Multiple Access



Exercise: (R)BTMA

We assume that eight mobile stations, numbered 1 to 8, are arranged in a line such that each station's reception range only reaches its immediate neighbors. We further assume that communication occurs according to the pattern 1→2, 3→4, 6→5, 8→7.

- What happens if the stations use BTMA or RBTMA, respectively?

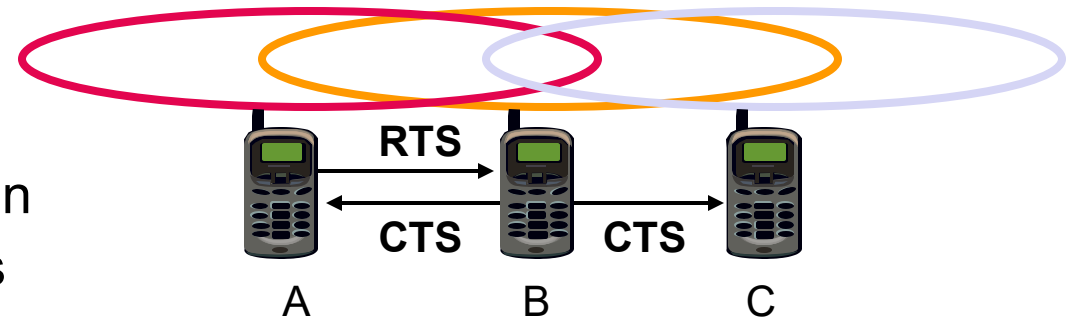


Approaches for Collision Avoidance (2)

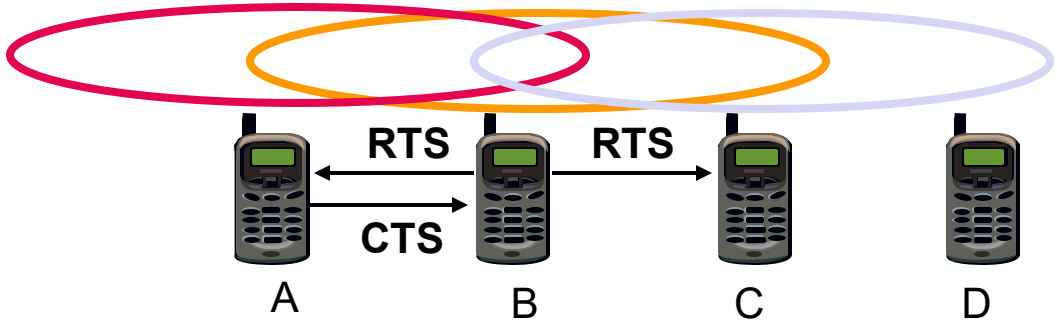
- Collision avoidance using in-band signaling
  - Control and data messages separated by time multiplexing
    - Control messages on the same frequency band as data transmission
- Example
  - Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)
    - Three-way handshake minimizes the number of hidden stations
    - Sender transmits short Request to Send (RTS) message
    - Receiver responds with Clear to Send (CTS) message
    - Sender transmits data
  - Collisions can still occur during the handshaking phase!
    - But with a lower probability, as RTS/CTS messages are very small

Example: RTS-CTS handshake

- Avoiding <u>hidden terminal</u>
  - A and C want to transmit to B
  - A transmits RTS (before C)
  - B answers with CTS
    - Contains communication duration
  - C defers transmission as it receives CTS from B



- Avoiding <u>exposed terminal</u>
  - B wants to transmit to A, C to D
  - C receives RTS but no CTS
  - C does not defer the transmission unnecessarily, as it does not receive CTS from A



Exercise: CDMA/CA

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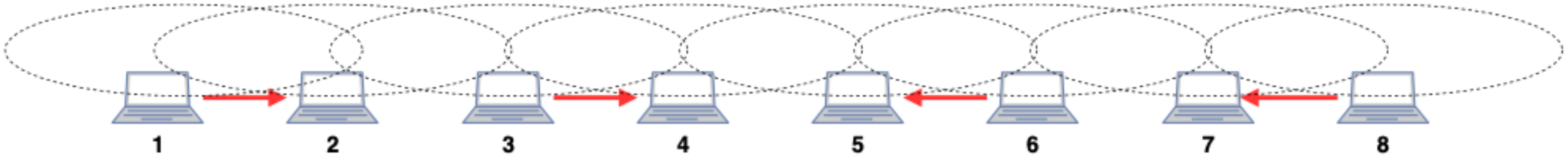


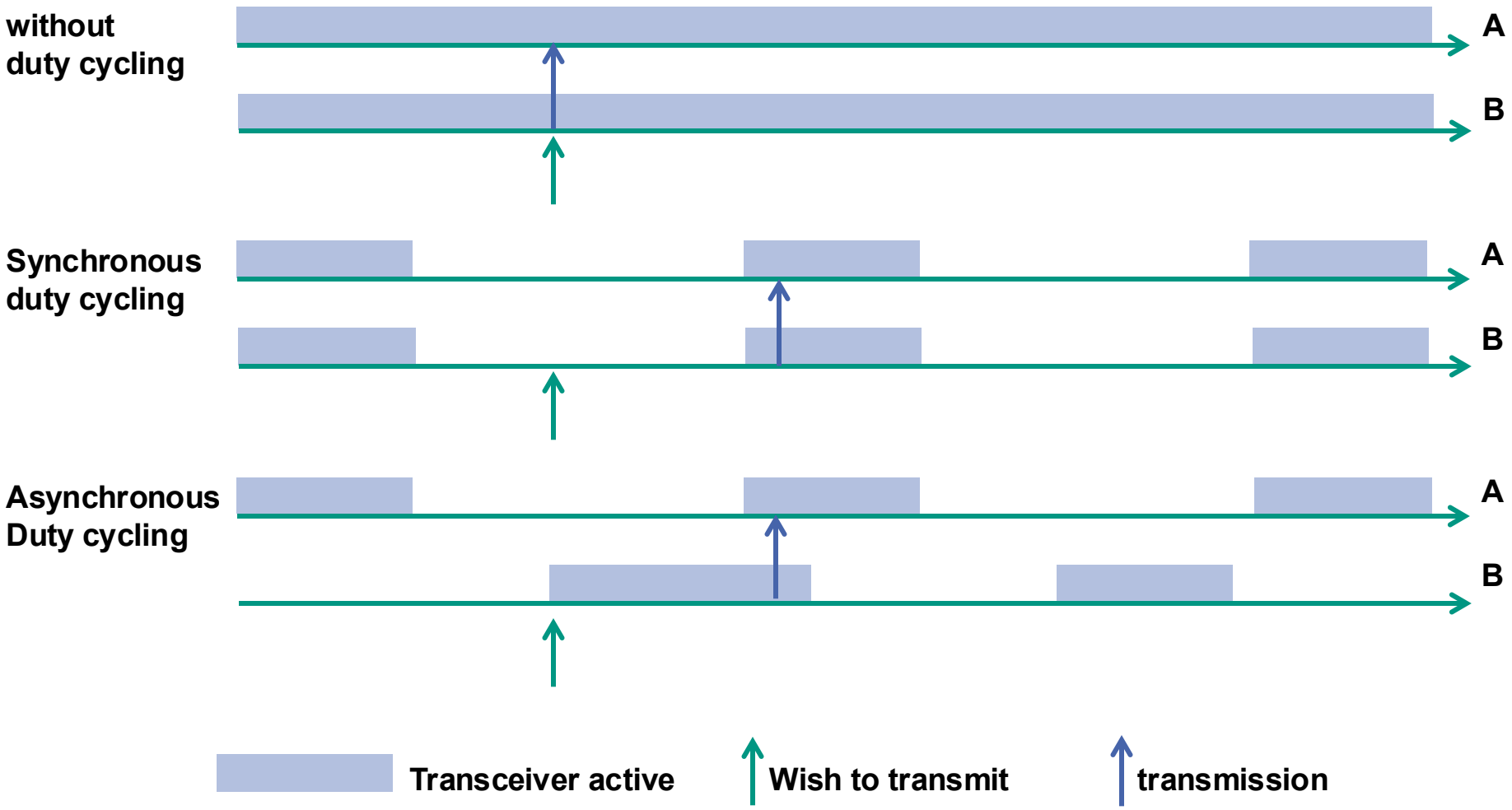
Energy Considerations

Duty Cycling

- Reasons for wasting energy
  - Collisions: Multiple stations transmit at the same time
  - Idle listening: Station waits for reception although no one transmits a packet
  - Overhearing: Station receives packet that is not targeted towards that device

- Solution: Duty Cycling
  - Deactivate transceiver as often as possible
    → sleep state (sometimes also called standby)
  - In order to exchange data receiver and transmitter must not be in sleep state
    → activate transceiver every once in a while
  - Reduces overhearing and idle listening but increases communication latency
  - Synchronous vs. asynchronous duty-cycling

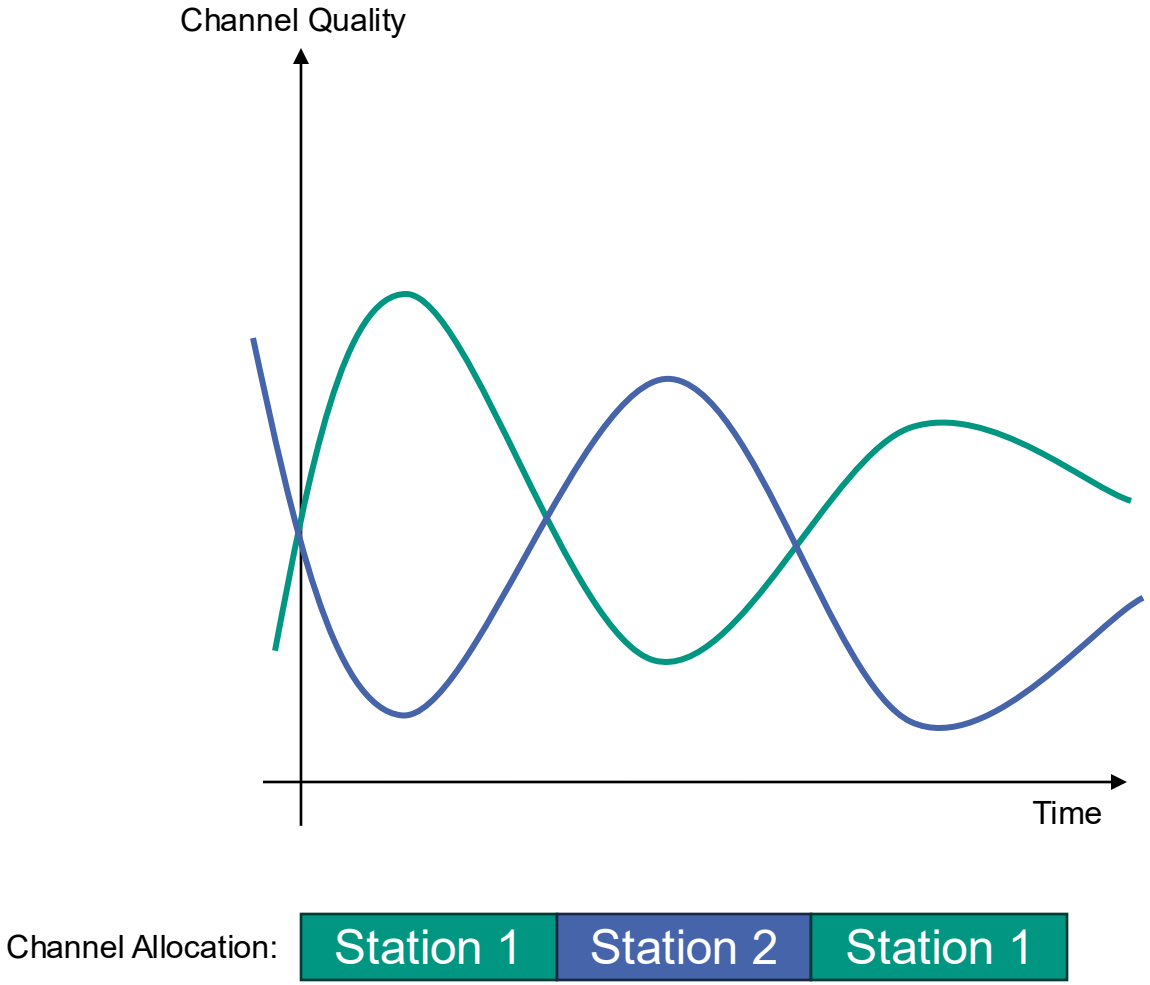
Example: Transmission B → A



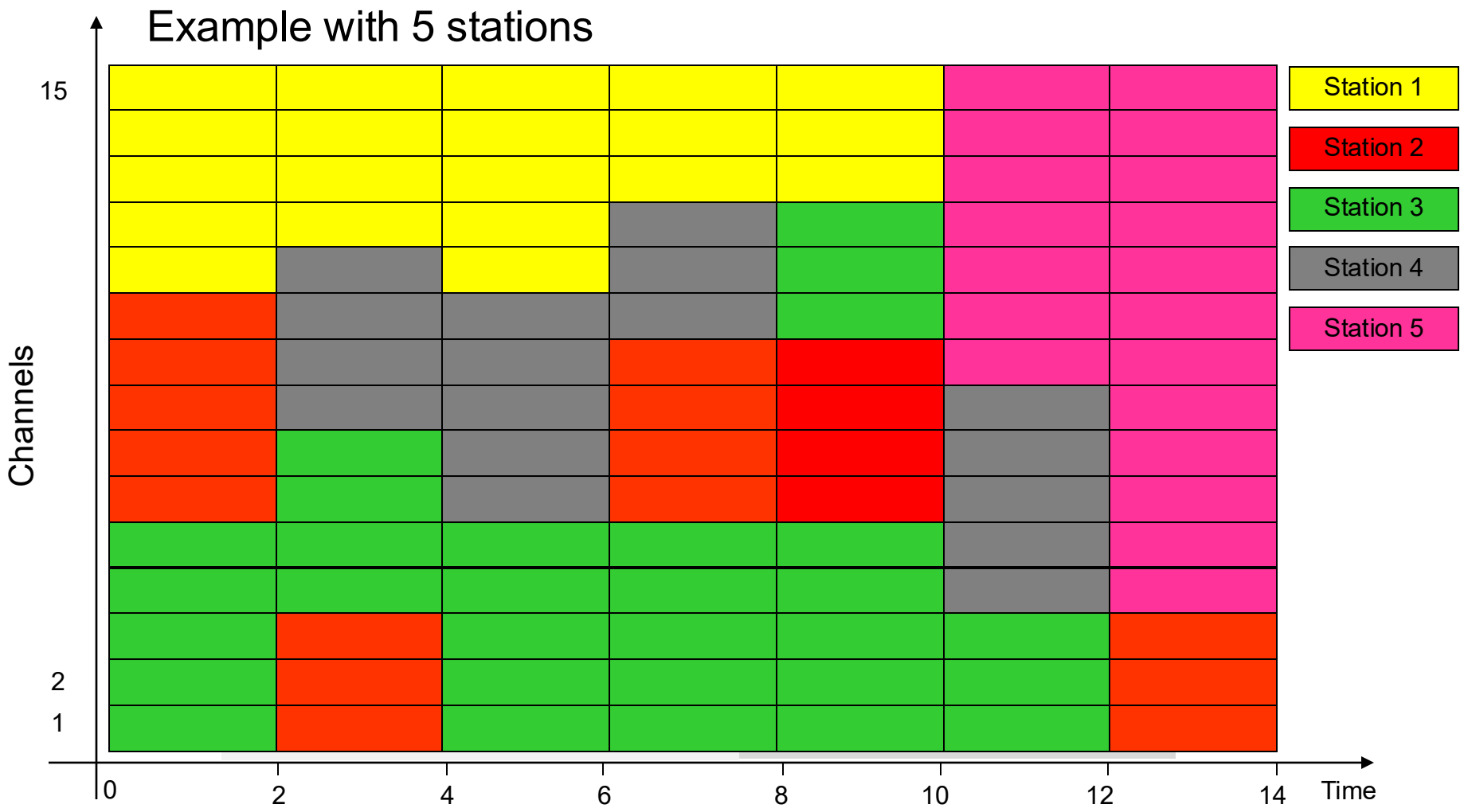
Centralized Controlled Access with Flexible Channel Allocation

Flexible channel allocation: Idea

- Observation: A station may not be able to fully utilize a centrally assigned channel due to fading
- But: Fading is helpful in multi-user environments!
  - Mostly uncorrelated for different stations
    → Multi-user diversity
  - Assign channel to station with the best channel quality
    → Throughput maximized
  - Gets better with increasing number of stations



Flexible channel allocation: Example



Flexible Channel Allocation: Scheduling

- Best Effort: Choose station with best channel quality
  - Advantages
    - Easy to implement and highest throughput
  - Disadvantages
    - Starvation in users with low channel quality

- Round Robin: Calling each station in turn
  - Advantages
    - Fair time allocation (with the same length of time slots)
    - Very simple approach
  - Disadvantages
    - Potentially low overall throughput: Channel may be assigned to station with poor channel quality

- Proportional Fairness: Choose station i with highest $R(t)/\overline{T}_i(t)$ at time t
  - $R(t)$: data rate based on channel quality, $\overline{T}_i(t)$: average long term throughput of station i
  - Advantages
    - Higher throughput than round robin if stations can be flexibly chosen based on channel quality
  - Disadvantages
    - May decrease overall throughput significantly

Summary

- Access to shared wireless medium is an important problem in mobile communications
  - Various multiplexing techniques in use

- Random access without carrier sensing: Aloha / Slotted Aloha
  - Basic simple procedure
  - Interesting in combination with reservation procedures
    - Collision only during the reservation phased, not during actual data transfer

- Random access with carrier sensing: (RI-)BTMA / CSMA/CA
  - Must cope with hidden and exposed stations
  - Busy tone
    - Requires second frequency band
  - RTS/CTS handshake
    - Collisions of control messages possible

- Duty cycling save energy but complicates medium access

- Flexible channel allocation can increase throughput in multi-user scenarios

Exercises - Media access

- 3.1 Why is CSMA/CD not used in wireless networks?
- 3.2 Name different types of duplex links.
- 3.3 What are the advantages and disadvantages of frequency division duplex and time division duplex?
- 3.4 How can media access protocols be classified?
- 3.5 What is the advantage of slotted aloha over "pure" aloha?
- 3.6 What is the advantage of splitting into reservation and data time slots in Reservation Aloha?
- 3.7 Explain the difference between implicit and explicit reservation.
- 3.8 How can the problems of hidden or delivered end devices be countered?

References and further reading

- [3.1] J. Schiller; Mobile Communications; Addison-Wesley, 2003
  Chapter 3 - Media access procedures
- [3.2] A. Chandra, V. Gummalla, J. Limb, Wireless Medium Access Control Protocols, IEEE Communications Surveys & Tutorials, www.comsoc.org/livepubs/surveys/public/2q00issue/gummalla.html, 6/2000
- [3.3] I. Stojmenovic, ed., Handbook of Wireless Networks and Mobile Computing - Chapter 6: Wireless Media Access Control, John Wiley & Sons, February 2002